Test Plan: VWO Login Dashboard

Objective

Verify that the VWO login experience provides secure, accessible, responsive, and performant access to the VWO platform, and that authentication and recovery flows behave as defined in the PRD.

Scope

Validate the login page and its related authentication, account recovery, session, enterprise access, and post-login transitions. Confirm that measurable targets in the PRD are assessed and that failures can be recovered from with clear user feedback.

Inclusions

- Email/password login, field validation, password requirements, and authentication error handling.
- Remember Me behavior, session handling, session timeout, and successful dashboard transition.
- Password reset and email-based account recovery.
- Optional MFA and enterprise SSO, subject to availability in the test environment.
- Registration/free-trial link and the login-page announcement, including light and dark themes.
- Responsive layout, auto-focus, clickable labels, loading states, keyboard navigation, screen-reader support, high contrast, and WCAG 2.1 AA checks.
- HTTPS enforcement, authentication data protection, secure session handling, rate limiting, and applicable GDPR/CCPA and enterprise audit requirements.
- Login-page load time, concurrent login behavior, availability, and geographic performance requirements.
- Core platform transition, login success/failure analytics, support integration, and any enabled third-party identity providers.
- Verification of the PRD's stated success metrics where suitable measurement data is available.

Exclusions and Conditional Scope

- Biometric authentication, adaptive authentication, and progressive web app capabilities are listed as future enhancements and are excluded from this release's acceptance testing.
- Social login is described as optional. Include it only if the release explicitly enables it and provides configured identity-provider test accounts.
- Optional MFA, SSO protocols, recovery methods, and integrations are tested only when implemented and configured for the release. Their exact supported modes must be confirmed before execution.

Test Environments

Use an approved, production-like non-production environment with HTTPS and the authentication, platform, email-recovery, analytics, support, and identity-provider integrations enabled for the release. Use dedicated test accounts representing standard and enterprise users, including accounts configured for MFA/SSO where supported.

The PRD does not specify browser/OS support, device matrix, environment URLs, identity-provider configuration, test data, or access to production-like load infrastructure. The project team must confirm these before execution. Do not use real user credentials or personal data in test cases.

Defect Reporting Procedure

Record each defect in the project-approved tracking system. Include a concise title, environment and browser/device, preconditions and test account type, reproducible steps, expected and actual results, evidence with sensitive data removed, severity/priority, and the affected requirement or test case. Triage security and privacy issues through the approved restricted process. Retest fixes in the originating environment and run relevant regression checks.

The tracker, severity definitions, response targets, and escalation contacts are not specified in the PRD and must be confirmed by the project team.

Test Strategy

- 1. Requirements review: Map each functional and non-functional requirement to one or more test cases. Resolve ambiguities before treating optional capabilities as release blockers.
- 2. Functional testing: Exercise valid and invalid login, field validation, recovery, remember-me, sessions, themes, registration navigation, loading, and success/error states.
- 3. Integration testing: Verify the authenticated transition to the VWO dashboard and behavior of enabled analytics, support, SSO, MFA, email, and social identity integrations.
- 4. Compatibility and accessibility: Test the confirmed browser/device matrix at mobile and desktop sizes; check keyboard-only use, focus order/visibility, labels, screen-reader announcements, contrast/high-contrast behavior, and WCAG 2.1 AA criteria.
- 5. Security testing: Validate security requirements using approved test accounts and authorized test environments. Include rate-limit behavior and session protections; coordinate deeper security assessment with the authorized security team. Do not conduct destructive or out-of-scope testing.
- 6. Performance and reliability: Measure login-page load against the 2-second target on the agreed standard connection, exercise the agreed concurrent-login profile, and verify monitoring/evidence for availability and geographic performance targets.
- 7. Regression testing: Rerun affected login, recovery, session, integration, accessibility, and responsive cases after relevant changes.
- 8. Metrics validation: Confirm how login success rate, satisfaction, support volume, uptime, and security outcomes are measured. Report observed values against PRD targets when reliable measurement data exists; do not infer a metric from a small functional test sample.

Test Schedule

The PRD provides implementation phases but no dates or estimates. Execute testing in these release-aligned stages:

Stage	Focus	Suggested timing
Phase 1: Core Authentication	Login, validation, errors, password reset, and core security checks	After core authentication is testable
Phase 2: Enhanced UX	Responsive behavior, accessibility, themes, loading states, and advanced validation	After the enhanced UX build is available
Phase 3: Enterprise Features	Configured SSO/MFA, security features, analytics, monitoring, and integration checks	After enterprise integrations are available
Release regression	Recheck critical flows and resolved defects across the agreed matrix	Before release approval

Dates, durations, owners, and environment availability are TBD by the project team.

Test Deliverables

- Requirement-to-test traceability and the test cases/results.
- Test data and environment readiness record, excluding secrets and personal data.
- Defect log with retest outcomes.
- Functional, compatibility, accessibility, security-coordination, and performance summaries.
- Release test summary documenting pass/fail/blocked status, unresolved risks, metric evidence, and approval decision.

Entry and Exit Criteria

Entry Criteria

- The release build and approved test environment are available over HTTPS.
- Requirements, supported browser/device matrix, enabled integrations, and optional feature scope are confirmed.
- Test accounts and safe test data are available, including configured MFA/SSO accounts where applicable.
- Critical integrations and test monitoring are available, and the defect reporting process is agreed.
- No known blocker prevents execution of the planned critical paths.

Exit Criteria

- All in-scope test cases have a recorded result; blocked or deferred tests have an owner and rationale.
- No unresolved critical or high-severity defects remain unless explicitly accepted by the release approver.
- Critical login, recovery, session, and dashboard-transition flows pass on the agreed supported matrix.
- Accessibility, security, performance, and reliability findings are documented and dispositioned by their responsible owners.
- Available measurements are compared with PRD targets: 95%+ login success rate, login page load within 2 seconds, 99.9% uptime, 90%+ login-experience satisfaction, 20% reduction in login-related support tickets, and stated security/compliance goals. Measurement periods, sources, and acceptance ownership must be agreed; the PRD does not define them.
- Test summary and residual risks are reviewed for release approval.

Test Execution

Execution Entry Criteria

- Entry criteria above are met for the test stage.
- The build/version, environment, and test data are recorded.
- Testers have the required approved access and know the escalation route for security/privacy issues.

Execution Exit Criteria

- Results and evidence are recorded for executed cases.
- Defects are filed, prioritized, and linked to affected requirements; fixes are retested.
- Failed, blocked, and not-run cases are clearly distinguished and reported.
- Regression for changed areas is complete or explicitly identified as outstanding.

Test Closure

Closure Entry Criteria

- Planned execution is complete or remaining work has an agreed disposition.
- Defect retesting and the release regression pass are complete.

Closure Exit Criteria

- The final test summary, traceability, defect status, and residual risks are delivered.
- Outstanding deviations from PRD targets are documented with an owner and decision.
- Required product, engineering, security, and accessibility approvals are recorded according to the project process.

Tools

- Playwright with TypeScript for browser-based functional and responsive checks, consistent with the skill metadata.
- Browser/device and screen-reader tools selected to match the confirmed support matrix.

- Approved performance, security, analytics, and defect-tracking tools are TBD; the PRD does not name specific products.

Risks and Mitigations

Risk	Mitigation
Optional MFA, SSO, social login, and recovery behavior are underspecified or unavailable.	Confirm release scope, protocols, providers, and test accounts before execution; report unavailable coverage as blocked, not passed.
Security tests or account lockout/rate limiting could affect shared systems or users.	Run only approved checks in isolated test environments with dedicated accounts; coordinate with the authorized security team.
Performance and availability targets lack workload profiles, measurement windows, and environment definitions.	Agree on the standard connection, concurrency profile, regions, and measurement method before benchmarking.
GDPR/CCPA, audit, and enterprise policy expectations are not mapped to specific controls.	Obtain compliance/security acceptance criteria and evidence requirements from the responsible owners.
The 95% login success, 90% satisfaction, and 20% support-ticket reduction targets depend on aggregate data outside a test run.	Define analytics sources and reporting periods; validate instrumentation and report actual trends separately from functional pass/fail.
Browser/device support and defect triage rules are absent from the PRD.	Confirm the matrix and project defect workflow before test execution.

Approvals

Role	Name	Decision	Date
Product owner	TBD	TBD	TBD
Engineering owner	TBD	TBD	TBD
QA/test owner	TBD	TBD	TBD
Security/compliance owner	TBD	TBD	TBD
Accessibility owner	TBD	TBD	TBD

Core Test Scenarios

ID	Area	Scenario and expected result
AUTH-01	Login	Submit valid email and password; authentication succeeds and the user reaches the VWO dashboard.
AUTH-02	Login	Submit incorrect credentials; access is denied and a clear, actionable error is shown without exposing sensitive information.
AUTH-03	Validation	Blur an empty or malformed email field; immediate, understandable validation feedback is shown and submission behavior is correct.

ID	Area	Scenario and expected result
AUTH-04	Validation	Submit missing or invalid required fields; invalid data is not accepted and field feedback is accessible.
AUTH-05	Password	Verify password requirement/strength feedback against the confirmed password policy; the UI does not claim requirements that differ from server enforcement.
AUTH-06	Loading	Submit login and observe the processing state; feedback is visible and duplicate submissions are handled safely.
AUTH-07	Remember Me	Log in with the option selected and unselected; persistence matches the defined session policy and does not exceed configured timeout/security rules.
AUTH-08	Session	Verify configured timeout and session-token behavior, including expired-session handling and return to an appropriate authentication state.
AUTH-09	MFA	For accounts with MFA enabled, complete valid and invalid second-factor flows; access is granted only after successful verification.
AUTH-10	SSO	For each enabled enterprise protocol/provider, verify successful sign-in, cancellation, and failure recovery without unintended access.
REC-01	Password recovery	Request reset for an eligible account; the user receives the expected recovery flow and can set a password meeting the confirmed policy.
REC-02	Password recovery	Exercise invalid, expired, reused, or otherwise unusable reset tokens; reset is denied and the user receives safe recovery guidance.
REC-03	Password recovery	Compare responses for eligible and ineligible account identifiers to verify the approved privacy behavior and avoid unnecessary account disclosure.
UX-01	Navigation	Follow the free-trial/registration link; it opens the intended registration destination.
UX-02	Branding/theme	Verify VWO branding, announcement content, and light/dark theme behavior at supported viewports.
UX-03	Responsive	Use the login page at agreed mobile and desktop viewport sizes; controls remain readable, usable, and do not overlap.
UX-04	Focus/labels	Verify initial focus, clickable labels, logical keyboard order, visible focus, and operation without a pointer.
UX-05	Screen reader	Verify accessible names, field errors, status/loading feedback, and interactive controls with supported assistive technology.
UX-06	Contrast	Check normal and high-contrast modes against WCAG 2.1 AA criteria for the in-scope page and states.
INT-01	Platform	Verify successful authentication transitions to the correct VWO dashboard and preserves the intended context from the prior session where specified.
INT-02	Analytics	Verify successful and failed login events are recorded as intended without sending passwords, tokens, or other authentication secrets.
INT-03	Support	Verify the login-assistance path reaches the configured customer support experience.
SEC-01	Transport	Verify login and recovery communications use HTTPS and that authentication data is not transmitted over insecure connections.
SEC-02	Rate limiting	Verify repeated authentication attempts are throttled according to the approved policy and legitimate recovery is possible.
SEC-03	Session security	Verify session tokens are generated, handled, and invalidated according to approved session requirements.
SEC-04	Data handling	Verify credentials are not exposed in page content, browser-visible logs, analytics, or error messages; validate stored-password controls through approved evidence/review.
PERF-01	Load time	Measure login-page load under the agreed standard connection; compare results with the PRD's 2-second target.
PERF-02	Concurrent access	Exercise the agreed concurrent-login profile and check response behavior, errors, and monitoring against the approved expectations.

ID	Area	Scenario and expected result
REL-01	Availability	Review service monitoring over the agreed observation window against the 99.9% uptime target.
REL-02	Geographic performance	Measure login experience from agreed regions and review CDN/global deployment behavior against defined expectations.

Open Decisions

- Confirm supported browsers, operating systems, devices, and screen readers.
- Define exact email/password validation and password-strength rules, authentication error wording, lockout behavior, session timeout, and remember-me persistence.
- Confirm supported MFA factors, SSO protocols/providers, social login providers, and password recovery methods for the release.
- Define reset-token lifetime and reuse policy, rate-limit thresholds, session-token requirements, and security evidence owners.
- Agree on performance load profiles, global regions, uptime measurement window, standard network conditions, and source of success/satisfaction/support metrics.
- Confirm compliance control mapping, test-data handling, defect workflow, dates, owners, and approvers.